Second Self

A interactive plateform based on an augmented mirror

2021,

> Second Self is an augmented reality platform developed for an augmented mirror, a mirror on which we can display information over the reflection. It can interact with the user using a color and depth camera combined with pose estimation. The goal is to create a platform hosting multiple modules for fitness, dance, sign language, cosmetics, and more.

The use of digital technology within homes and offices is increasing, while the latest technological and digital breakthroughs allow significant advancement in this field. Better sensing and recognition techniques are now available, allowing the developed solutions to interact with the environment and the user.

One omnipresent component of homes and public spaces is mirrors. Different sorts of mirrors integrating recent technology currently exist. Magic mirrors are the most common type of augmented mirror, especially since they can easily be built at home. It is possible to display information on top of the reflection, such as the weather, time, or current traffic situation using a one-way mirror and a screen,

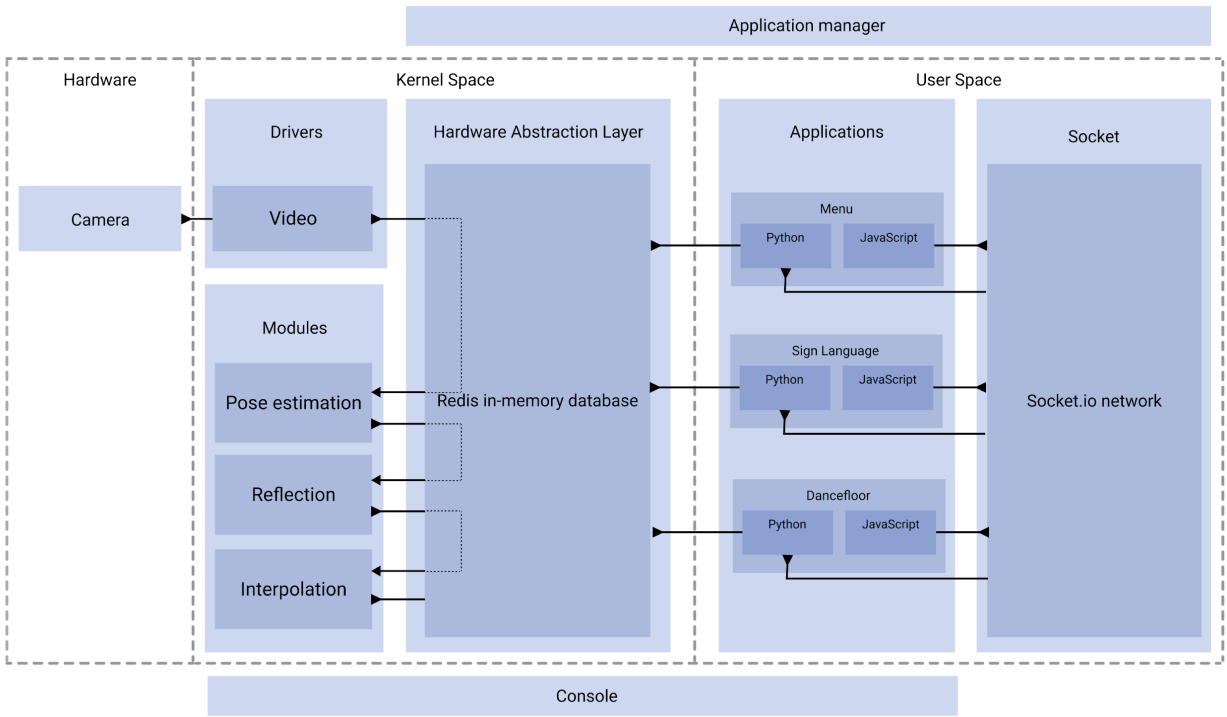
The technologies integrated into such mirrors are far from reaching the current possibilities of the digital world. The interactivity and immersion that could be obtained are not exploited fully.

Therefore, we propose a new type of mirror, the augmented mirror, a platform that provides extensive interaction between the real and the virtual world. The objective is to create a platform for recreational, medical, and educational use.

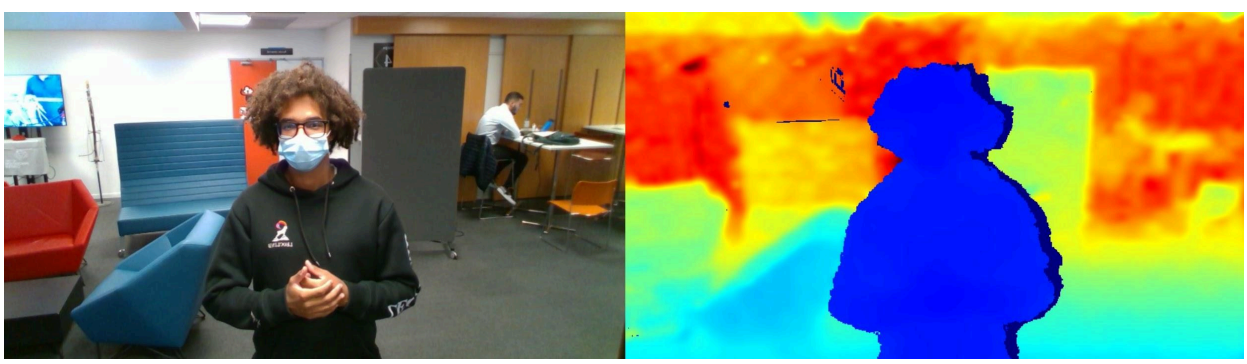
Platform Architecture

The hardware of this mirror follows the same principle as other similar augmented reality mirrors. A one-way mirror is placed against a screen. The mirror reflects perfectly where the screen is black and can display information when the pixels are emitting light. A camera is placed at the top of the mirror, facing slightly down. A laptop is placed at the back of the screen. We chose a laptop to reduce the overall size of the mirror.

The mirror uses GOSAI to power its applications. Several drivers and modules were developed while creating this platform.



For the platform to understand the user's position and the environment around the user, it is necessary to use a camera that captures not only regular color frames but also depth. This allows applications to scan the room or detect and position objects the user interacts with. We thus use the Intel D435 camera. It is a webcam with an infrared light emitter, two infrared sensors, and a regular color camera. The two infrared cameras use the same process as the brain to evaluate the distance: they simultaneously capture a frame and process these frames to produce a single frame where each pixel contains a depth value. It is then possible to align the depth frame with the color frame to have a pseudo-3D representation of the environment.



A video driver was then created to process the frames and send the data to the in-memory database. It supports both a regular webcam as well as an intel depth and color camera. The resolution of the video as well as the frame rate can be configured by the user.

To detect the user's position and adjust the elements displayed based on it, a pose estimation module was created. It requires the color frame from the video driver. When an RGB image is captured by the video driver, the module processes it using the pose estimation solution from mediapipe. This model was selected for multiple reasons: it is easy to implement, other detection models such as iris detection or hair segmentation from mediapipe can be implemented in the exact same way with very few differences, and lastly, it is fast (the processing time is similar to the time between two frames of the camera) while providing a good accuracy. On average, the camera provides a frame every 33ms, and the pose estimation module takes between 25 and 35 ms to process the frame.

The result from the pose estimation is expressed in a unit relative to the camera. It is necessary to have the world position of the user relative to the mirror to align these points with the reflection. Another module was thus created with this goal: project pixel coordinates into reflection coordinates using the user's eyes position, the camera's intrinsic parameters and a depth frame. This process involves simple mathematical trigonometry operations.

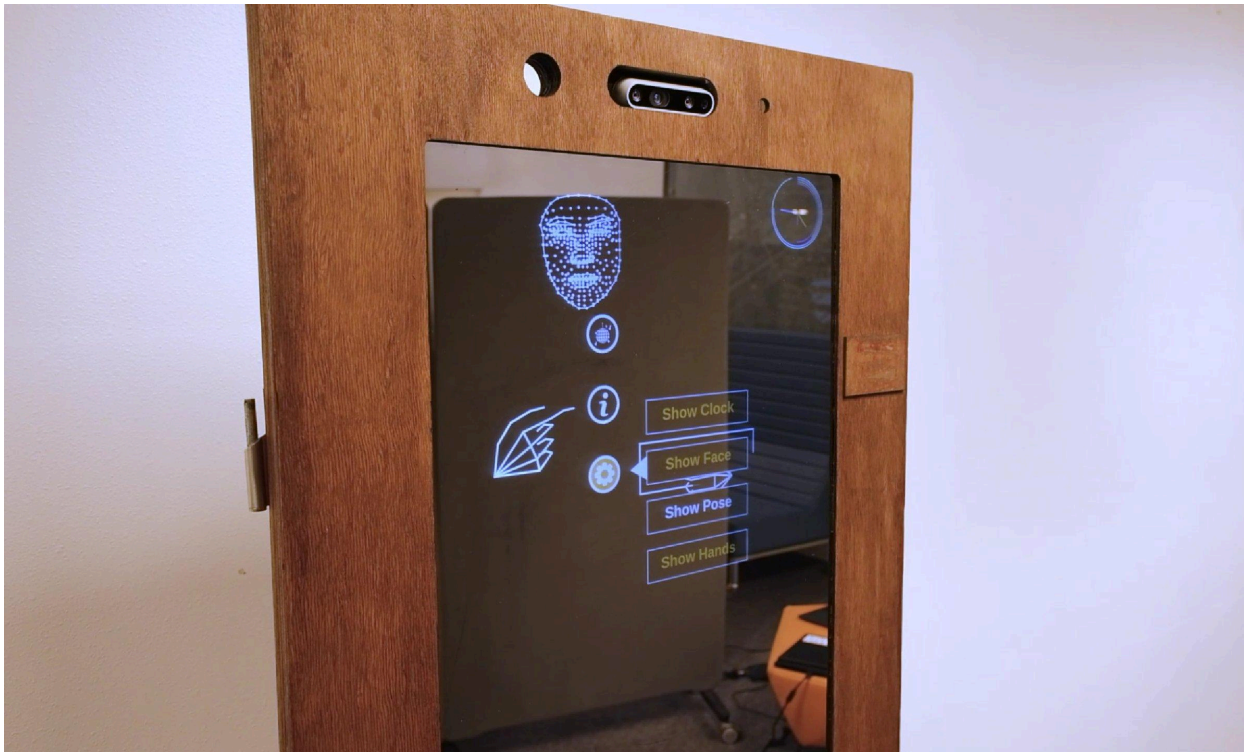
Finally, a module is used to interpolate between the points to improve the fluidity of the movements.

The final data is then provided to multiple applications developed on the User Interface side.

Applications Examples

Menu

The first application developed on this platform is a menu. This menu allows a user standing in front of the mirror to launch other applications and to have basic control over the mirror. This application uses the pose estimation module to detect the user's position and to allow the user to control the mirror.



A series of icons and boxes are linked to actions. Three sections are available:

- A sub-menu to start and stop applications.
- A panel describing the pannel in short sentences.
- A sub-menu to hide or show the overlay of different parts of the pose estimation

To open a section, the user must hold the tip of the left index acting as a cursor over the circular button. It can be difficult to do when a problem of alignment occurs, in which case the cursor is not aligned with the real position of the user's finger. Once the cursor is in the right area, a selection animation is shown. At the end of the animation, the section is opened. In the same way, to launch an application from the sub-menu, the user must hold the tip of the right index over the box with the name of the application.

Two ways of using the menu were studied. The first one is to link the root position of the menu with the left hand of the user and to open it when a gesture is performed. This allows the user to move both hands, offering a more extensive control over the process of selection of an element. This method improved the immersion of the user and offered the same experience for all users but showed disadvantages in terms of practicality and ease of use.

Another approach was to lock the menu in place and open it when the user holds the reflection of the tip of their right hand over a circular button. This method allowed the user to only use

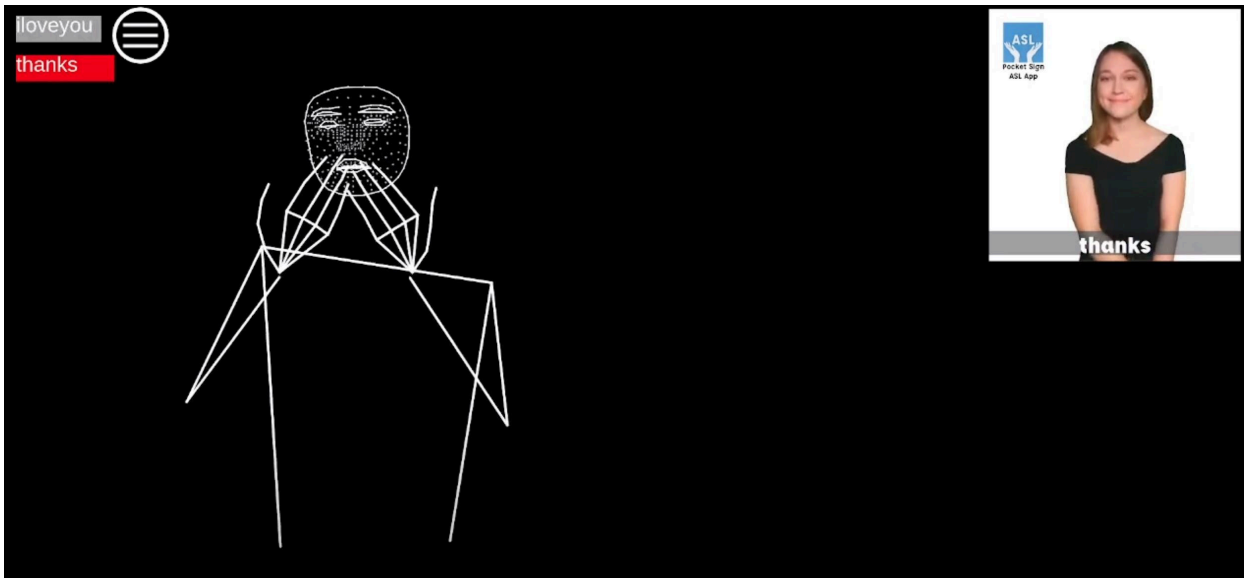
one hand to navigate and interact. However, the position of this button and the position of the menu proved to have a big impact on the experience of the users. One position can prove to be comfortable for a user but uncomfortable for another.

The menu is the only application launched by default. Without it, the user would have to launch the application via the control or dashboard pages from another device.

Sign Language

Learning Sign Language is a challenging task. An application was developed on this platform by Adrien Lefevre, another student from the DVIC, to improve the way learning Sign Language is done. The objective of this application is to introduce the user to sign language and learn how to use it by seeing a series of signs and trying to reproduce them.

The Sign Language Recognition (SLR) module was first implemented as a new module for GOSAI. It uses the data from the pose estimation module, and with the help of a neural network, interprets the data and produces a sequence of signs. Thirty frames of pose estimation data are used to produce an estimation of a sign. A total of 12 different signs can be recognized, and more will be added in the future.



Once the SLR is performed, the data is sent to the frontend of the application. A video at the top-right of the screen shows the sign that needs to be reproduced. In the top left, the user can see the name of the sign that needs to be reproduced as well as the sign that the SLR module has recognized. The width of the bar behind the recognized sign is proportional to the confidence of the SLR module.

Dance Floor

Dance requires a lot of control over motions and postures. Practicing for hours and having a coach is important to quickly show improvements, and the training often comes to repeating the same movements over and over until mastery. However, having a coach around is not always possible, especially when the practice is done at home. This application was developed to offer a simple alternative for beginners to practice their dance while having a coach in the form of an application.

The Dance Floor application requires the pose estimation module.

The reference dance has to be pre-recorded and processed. We first took a video of a performer performing simple dance motions. Then, each frame of the video was processed by a pose estimation model: the keypoints of the performer were saved in a JSON file and the performer was segmented from the background to apply a mask and have a black background for the video. The video was then converted to GIF format and saved in the assets folder along with the JSON file.

When the user starts the application, a first initialization is performed. The user must stand still with their arms at the sides. This is done to scale the playback video to the size of the user and position it in alignment with the user's reflection. This way, the user can directly see how his body is positioned compared to the performer and correct his position accordingly. To have a direct idea of the user's performance, we added a bar on the side of the screen. The height of

the bar is a direct representation of the distance between the user's and the performer's body position: the smaller the bar is, the closer the user is to the performer. Under a certain threshold, the bar turns green and the video plays. Over the same threshold, the bar turns red and the video stops. A timer ensures that the movements are performed fluidly. Every time a movement is completed, time is added to the timer. Once the timer reaches zero, the dance application restarts.

The format of the video is GIF because it seems to be the only way to play a video frame by frame on the web. This has the drawback of adding a ~10 seconds loading time when the application is launched for the first time.

Discussions

This platform offers a lot of possibilities in terms of applications, so much so that it is easy to forget the existing ones. The menu in itself could be done in a multitude of ways, and every aspect deserves to be explored and researched. A deeper understanding of the most adapted ways of interacting with the mirror is needed to improve the experience of the user. The usual reliance on touch makes this platform often unintuitive to use. Simple gesture interactions were explored, but other methods could be employed like voice recognition, the use of a controller or even eye tracking.

The information displayed on the mirror is a key aspect of the experience. The user must be able to see the information he needs without having to make too much effort and the placement of the information needs to be coherent. This information must be relevant and not too intrusive. Using the current version of the mirror tends to indicate that it is easy for a user to forget the reflection and focus their attention solely on the information displayed on the mirror. Finding a balance between the information displayed and the reflection is a challenge that needs to be addressed.

Different applications do not require the same setup. For example, the current screen size is perfect to practice sign language but isn't wide enough to fully display the user when doing wide movements in the dance application, reducing the overall experience and the usability of the application. Defining the requirements for each application is a challenge, and it is important to have a clear understanding of the user's needs and possibilities in terms of hardware...